Darshan UNIVERSITY
योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 7

Roll No.: 514

Name: Meet Indoriya

01) WAP to find sum of tuple elements.

Sample Input:

```
T = (10, 20, 30, 40)
```

Sample Output:

```
Sum of tuple elements = 100
```

In [1]:
```
T1 = (10, 20, 30, 40)
sumT1 = sum(T1)
print("Sum Of Tuple Elements:",sumT1)
```

```
Sum Of Tuple Elements: 100
```

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

```
T = (7, 2, 9, 4, 1, 5)
K = 2
```

Sample Output:

```
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

In [2]:
```
T2 = (7, 2, 9, 4, 1, 5)
K = 2
x = sorted(T2)
minimum1 = x[:K]
maximum1 = x[-K:]

print(minimum1)
print(maximum1)
```

```
[1, 2]
[7, 9]
```

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

```
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

In [3]:
```
T3 = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]

print("The original tuple list is : " + str(T3))
K = 2

l1 = []
for i in T3:
    count=0
    for j in i:
        if(j%K==0):
            count+=1
    if(count==len(i)):
        l1.append(i)
print("Tuples divisible by 2 :",str(l1))
```

```
The original tuple list is : [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3,
27), (4, 64), (5, 125)]
```

In [4]:
```
T4 = 1,2,3,4,5
newT4 = [(i,i**2) for i in T4 ]
print("List of tuples with number and its cube:",newT4)
```

```
List of tuples with number and its cube: [(1, 1), (2, 4), (3, 9), (4, 16), (5, 2
5)]
```

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3,
-4, -5)]
```

Sample Output:

```
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

In [5]:
```
T5 = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
l2 = []
for i in T5:
    numCount = 0
    for j in i:
        if j > 0:
            numCount += 1
    if numCount == 3:
        l2.append(i)
newT5 = tuple(l2)
print("Tuples with all positive elements:",newT5)
```

```
Tuples with all positive elements: ((1, 2, 3), (4, 5, 6))
```

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

```
List after removing tuples of length 2 : [(3, 4, 5), (6,), (7,
8, 9)]
```

In [6]:
```
T6 = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
k6 = int(input("Enter The value of K"))
l6 = []
for i in T6:
    if len(i) != k6:
        l6.append(i)
print(l6)
```

```
[(3, 4, 5), (6,), (7, 8, 9)]
```

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

```
(5, 7, 9)
```

In [7]:
```
T7 = (3, 5, 3, 7, 3, 9)
x = int(input('Enter The X value: '))
l7 = list(T7)
print(l7)
l7 = [item for item in l7 if item != x]
newT7 = tuple(l7)
print("New Tuple elements: ",newT7)
```

```
[3, 5, 3, 7, 3, 9]
New Tuple elements:  (5, 7, 9)
```

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

```
(3, 5, 7, 9)
```

In [8]:
```
T8 = (3, 5, 3, 7, 3, 9)
print("The tuple elements are:",str(T8))
l8 = []
for i in T8:
```

```
    if i not in l8:
        l8.append(i)
newT8=tuple(l8)
print("The tuple after removing duplicates : ",str(newT8))
```

```
The original tuple is : (3, 5, 3, 7, 3, 9)
The tuple after removing duplicates :  (3, 5, 7, 9)
```

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

In [2]:
```
T9 = (2, 3, 4, 5)
print("The tuple elements are:",str(T9))
l9 = []

for i in range(len(T9) - 1):
    l9.append(T9[i] * T9[i+1])
l9 = tuple(l9)

print("Resultant tuple after multiplying adjacent elements:",str(l9))
```

```
The tuple elements are: (2, 3, 4, 5)
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

In [3]:
```
T10 = (1, 4, 5, 6)
T10_2 = (1, 2, 2, 3)
print("The original tuple is : " + str(T10))

li10 = len(set(T10)) == len(T10)
print("Is tuple distinct ? : " + str(li10))
```

```
The original tuple is : (1, 4, 5, 6)
Is tuple distinct ? : True
```

11) WAP to rotate the elements of a tuple to the right by `k` positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

In [3]:
```
T11 = (10, 20, 30, 40, 50)
k11 = 2

k11 = k11 % len(T)
rotated_tuple = T[-k11:] + T[:-k11]

print(f"Original Tuple: {T11}")
print(f"Rotated Tuple: {rotated_tuple}")
```

```
Original Tuple: (10, 20, 30, 40, 50)
Rotated Tuple: (40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

In [12]:
```
T12 = (1, 3, 5)
T12_2 = (2, 4, 6)

if len(T12) == len(T12_2):
    new_T12 = T12 + T12_2
    new_T12 = tuple(sorted(T12 + T12_2))

print(new_T12)
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

```
T = (4, 12, 7, 18, 10)
```

Sample Output:

```
(144, 324)
```

In [14]:
```
T13 = (4, 12, 7, 18, 10)
result = tuple(x**2 for x in T13 if x % 2 == 0 and x > 10)
print(result)
```

```
(144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

```
T = (4, 7, 9, 11, 15)
```

Sample Output:

```
((1, 7), (3, 11))
```

In [16]:
```
T14 = (4, 7, 9, 11, 15)

def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

result14 = tuple((i, v) for i, v in enumerate(T14) if is_prime(v))
print(result14)
```

```
((1, 7), (3, 11))
```

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

In [1]:
```
T15 = ('a', 'b', 'c', 'd', 'e')

evenTuple = T15[::2]
oddTuple = T15[1::2]

print("Even-index tuple:", evenTuple)
print("Odd-index tuple:", oddTuple)
```

```
Even-index tuple: ('a', 'c', 'e')
Odd-index tuple: ('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

In [6]:

```python
T16 = ([1, 2, 3], [4, 5, 6], [7, 8, 9])

newT16 = []
for i in range(len(T16[0])):
    s = 0
    for j in T16:
        s += j[i]
    newT16.append(s)

print(tuple(newT16))
```

```
(12, 15, 18)
```

In [ ]: